

Dylan MacDonald

Backend Software Engineer

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SKILLS

Languages	Python, C++, C#, SQL, JavaScript, Bash
Backend & Data	Django, Apache Kafka, MySQL, Google Cloud BigQuery, REST & RPC APIs
Frontend	React
Tools	Docker, Git, Jenkins CI/CD, Agile/Scrum, automated testing, AI-assisted development (Claude Code)
Engines	Unreal Engine, Godot

EXPERIENCE

Backend Software Engineer at A Thinking Ape

May 2026 – Present

- Designed and built a Kafka-based payment-processing service in Python/Django with a React admin panel that has decoded more than 4 million payment events, redesigning the underlying data model to guarantee correctness under out-of-order delivery.
- Built a subscription-conflict resolution system across 30+ microservices, with cancellation logic tailored to each payment platform, eliminating a recurring incident that had cost the team 18+ hours across 8 cases in six months.
- Authored the technical design document and led implementation of a first-party authentication service for the studio's web storefront, which serves roughly 13,000 monthly users and is the primary revenue channel for one of the company's live titles.
- Extended an existing BigQuery data pipeline with new match-telemetry streams to support in-game balance analysis.
- Wrote and reviewed code in Agile sprints, testing changes locally with Docker and AI-assisted tooling (Claude Code) before shipping them through the team's Jenkins CI/CD pipeline.

Junior Software Engineer at Offworld Defence Simulations

May 2021 – June 2022

- Built real-time multiplayer networking in C++ for military training simulations on Unreal Engine, synchronizing 100+ concurrent users with sub-millisecond update latency and minimal bandwidth overhead.
- Converted Blueprint prototypes into shipped C++ systems used by customers, including a networked training environment and supporting features for lighting and mine-plow simulation.
- Applied targeted performance optimizations across the simulation product to meet strict responsiveness requirements, contributing measurable gains across multiple systems.
- Mentored a Blueprint developer in C++ fundamentals through pair programming and debugging sessions, and was promoted from the company's first co-op position to full-time junior engineer after delivering the multiplayer networking systems.

Software Engineer Co-op at Intel Corporation

Sept 2020 – Apr 2021

- Modeled SSD controller chip performance in C++ to predict clock speed and inform architecture decisions (die size, power budget) ahead of fabrication.
- Automated benchmark simulation and data collection with Python and Bash, cutting engineer time spent processing results by roughly 80%.
- Wrote C++ validation tests comparing model predictions against real hardware measurements to isolate and correct modeling inaccuracies.

EDUCATION

B.Sc. in Computer Science, University of British Columbia, Vancouver, BC

2024